ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** dendrite	Dendrite tip receives the signal.
2	**(a)** phototropism	Positive phototropism.
3	**(a)** adrenaline	Adrenal glands.
4	**(b)** synapse	Chemical signal crosses it.
5	**(a)** gravity	Positive geotropism.
6	**(c)** pituitary	It controls other glands.
7	**(c)** insulin	Insulin is an animal hormone.
8	**(b)** thigmotropism	Response to touch.
9	**(a)** growth hormone	Pituitary.
10	**(c)** auxin	Auxin elongates cells on the shaded side.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflex is automatic and involuntary; controlled by spinal cord (1); Walking is voluntary and controlled by the brain (1).	1+1
12	Reflex is controlled by the spinal cord (1); The brain is informed afterwards and does not control the action (1).	1+1
13	Iodine is needed to make thyroxine (1); Prevents goitre (1).	1+1
14	Electrical impulse changes into a chemical signal (1); Chemical crosses the gap and starts a new impulse (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Receptor in the skin detects heat (1); Sensory neuron to spinal cord; relay neuron; motor neuron (1); Effector muscle withdraws the hand (1).	1+1+1
16	Insulin: pancreas, sugar level (1); Thyroxine: thyroid, metabolism (1); Adrenaline: adrenal, emergency (1).	1+1+1
17	Cerebrum: thinking (1); Cerebellum: balance (1); Medulla: involuntary actions (1).	1+1+1
	OR: Quick response to danger (1); Saves time as brain is not involved (1); Example: knee jerk or touching a hot plate (1).	1+1+1
18	Leaves fold when touched (1); Water leaves the cells of the pulvinus (1); Cells shrink and the leaf droops (1).	1+1+1
	OR: Hunger centre in the forebrain (1); Detects low blood sugar (1); Sends signals for eating (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dendrites receive information (1); Cell body (1); Axon carries impulse away (1); Nerve endings release chemical (1); Synapse transmits to next neuron (1).	1+1+1+1+1
	OR: Light causes auxin to shift to the shaded side (1); Cells on that side elongate more (1); Shoot bends towards light (1); Auxin made at shoot tip (1); Experiment with a covered tip shows no bending (1).	1+1+1+1+1
20	Receptor -> sensory neuron -> spinal cord (1); Relay neuron -> motor neuron -> effector (1); Reflex: involuntary and fast (1); Voluntary: brain involved and slower (1); Reflex examples: knee jerk (1).	1+1+1+1+1
	OR: Blood sugar rises after a meal (1); Pancreas releases insulin (1); Sugar is stored/used (1); Insulin release falls as sugar falls (1); Deficiency causes diabetes (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.

- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** chemotropism	Chemical stimulus.
2	**(d)** insulin	Blood sugar is high.
3	**(d)** thyroxine	Thyroxine regulates metabolism.
4	**(b)** change in turgor pressure	Water movement out of cells.
5	**(a)** iodine	Swollen neck due to thyroid.
6	**(c)** thyroxine	Iodine is needed for thyroxine.
7	**(a)** cytokinin	Found in fruits and seeds.
8	**(c)** cranium and fluid-filled balloon (meninges)	Skull and cerebrospinal fluid.
9	**(d)** pancreas	Beta cells of the pancreas.
10	**(d)** abscisic acid	Abscisic acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Auxin collects on the shaded side of the shoot (1); Cells there grow longer, bending the shoot toward light (1).	1+1
12	Hormones from endocrine glands (1); Carried by blood to target organs (1).	1+1
13	Cerebellum (1); In the hindbrain (1).	1+1
14	Signals between brain and body below the injury (1); Reflex arc and voluntary movement (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Forebrain: thinking, memory, hunger (1); Midbrain and hindbrain: reflexes of the eye (1); Cerebellum: balance; medulla: involuntary actions (1).	1+1+1
16	Hormone amount is regulated by the body (1); Example: insulin level depends on blood sugar (1); High sugar causes release of insulin (1).	1+1+1
17	Nervous: fast, electrical impulse (1); Hormonal: slow, chemical messengers via blood (1); Nervous effect short-lived; hormonal long-lasting (1).	1+1+1
	OR: Bony box: cranium (1); Fluid-filled balloon (meninges) (1); Shock absorption (1).	1+1+1
18	Touch-me-not: change in turgor, not growth (1); Tendril: differential growth (1); Both respond to touch (1).	1+1+1
	OR: Geotropism in roots (1); Positive geotropism (1); Shoots show negative geotropism (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Forebrain: cerebrum for thinking (1); Hunger and sensory areas (1); Midbrain: involuntary reflexes of eye (1); Hindbrain: medulla, pons, cerebellum (1); Cerebellum: balance and precision (1).	1+1+1+1+1
	OR: Speed (1); Type of signal (1); Duration (1); Pathway (1); Examples (1).	1+1+1+1+1
20	Phototropism (1); Geotropism (1); Hydrotropism and chemotropism (1); Thigmotropism: tendrils (1); Nastic: Mimosa leaves fold on touch (1).	1+1+1+1+1
	OR: Cranium protects brain (1); Vertebral column protects spinal cord (1); Meninges and fluid (1); Spinal cord is the reflex centre (1); Sends information to brain (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.

- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** receptor, sensory neuron, spinal cord, motor neuron, effector	Complete pathway.
2	**(c)** cerebellum	Hindbrain.
3	**(c)** gibberellin	Gibberellins and auxins.
4	**(c)** cerebrum	Forebrain.
5	**(d)** withdrawal of hand from a hot object	Automatic, quick, involuntary.
6	**(d)** tips of shoots and roots	Meristematic regions.
7	**(a)** dendrite -> cell body -> axon -> nerve ending	Direction of impulse.
8	**(b)** medulla in the hindbrain	Medulla.
9	**(c)** roots	Roots grow away from light.
10	**(d)** testes	Male sex hormone.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Their pancreas produces too little insulin (1); Insulin regulates blood sugar level (1).	1+1
12	Olfactory receptors in the nose detect smell (1); Information is sent to the forebrain (1).	1+1
13	Hormones are produced in the tip (1); They diffuse to the area needing them (1).	1+1
14	Detect stimuli from the environment (1); Ear (sound), eye (light) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Phototropism: shoot toward light (1); Geotropism: roots downward (1); Hydrotropism: roots toward water (1).	1+1+1
16	It secretes hormones that control other glands (1); Also secretes growth hormone (1); Located below the brain (1).	1+1+1
17	Secreted in fear or stress (1); Increases heart rate and breathing (1); Muscles get more blood (1).	1+1+1
	OR: Auxin: cell elongation (1); Cytokinin: cell division (1); Abscisic acid: inhibits growth (1).	1+1+1
18	Iodine: goitre (1); Growth hormone: dwarfism (1); Excess growth hormone: gigantism (1).	1+1+1
	OR: Promote cell division (1); Abundant in fruits and seeds (1); Delay leaf ageing (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Auxin: elongation, phototropism (1); Gibberellin: stem growth (1); Cytokinin: cell division (1); Abscisic acid: inhibits growth (1); Examples of location (1).	1+1+1+1+1
	OR: Touch: tendril coils by differential growth (1); Sensitive plant: turgor changes (1); Light: auxin (1); Water: hydrotropism (1); Gravity: roots grow down (1).	1+1+1+1+1
20	Pituitary: growth hormone, dwarfism (1); Thyroid: thyroxine, goitre (1); Pancreas: insulin, diabetes (1); Adrenal: adrenaline (1); Testes: testosterone; ovary: oestrogen (1).	1+1+1+1+1
	OR: Sensory receptors in skin (1); Impulse along sensory neuron (1); Spinal cord relays (1); Motor neuron to muscles (1); Brain is informed through other neurons (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.

- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND